

The Coin Is a Constant

Feedback as Detuning vs Shaping: the Measurable Split of the Demon's Action in Tilt Space

Michael Grafiel S Puno

August 2026

L.O.R.E. Framework (Law of Repulsive Emanation)

Abstract

Ch.80 found the feedback coin TRANSLATES the work-tilt curve: $a(t) = \langle W e^{(-tW)} \rangle / \langle e^{(-tW)} \rangle$ is antisymmetric about a SHIFTED center $c = a(1/2) = -0.073$ engaged, -0.129 harvest, the mirror moved but not bent. This chapter asks HOW MUCH of the coin is a pure constant shift. For an unconditional constant bias δ (every trajectory's work shifted by δ - the leading effect of a detuned control), the tilted mean obeys $a(t) \rightarrow a(t) + \delta$ at every t , so $c = \delta = \Delta\mu$. Measured: engaged $(0.35/2)$ $c = -0.0726$ vs $\Delta\mu = -0.0683$, $|c - \Delta\mu| = 0.004$ (6%) and a central reshaping residual $\text{netR} = 0.037$ - the coin is, to first order, a pure detuning. The stronger harvest $(0.05/16)$ deviates: $c - \Delta\mu = +0.025$ (19%) and $\text{netR} = 0.289$ (EIGHT times larger) - the coin is now partly a SHAPING, the state-dependence of the bit. The demon's action splits into a constant part (detuning, invisible to the mirror) and a conditional part (reshaping), and the split is measurable.

1. Detuning vs Shaping

A control that merely detunes the engine - adding a fixed offset to the accumulated work regardless of the trajectory - moves the distribution by a constant δ . Under the exponential tilt this means $a_{\text{detuned}}(t) = a(t) + \delta$, so its center (Ch.80) is shifted by exactly δ and nothing else changes: the mirror translates and never deforms. A control that CONDITIONS on the state, as the information engine does (Sagawa-Ueda 2010; Parrondo-Horowitz-Sagawa 2015), cannot be a single constant: it also reshapes.

2. The Measured Split

Same Heun SRK2, seed 42, rows of Ch.76-80:

row	$c = a(1/2)$	$\Delta\mu$	$c - \Delta\mu$	netR (central)
control	-0.0004	---	---	---
engaged	-0.0726	-0.0683	-0.0042 (6%)	0.037
harvest	-0.1294	-0.1547	+0.0253 (19%)	0.289 (8x)

For a pure constant detuning $c = \Delta\mu$ would hold EXACTLY and the reshaping residual netR (the mean $|a(t) - [a_{\text{control}}(t) + c]|$ over the well-conditioned central window t in $[-0.5, 2.0]$) would vanish. The engaged coin meets this to 6% with $\text{netR} = 0.037$; the harvest deviates to 19% and its reshaping residual is EIGHT times larger. The amount the coin departs from a translation is exactly the amount it is state-dependent.

3. The Mirror Cannot See the Constants

Ch.77-80 read the mirror $P(+W)/P(-W) = e^W$, its cumulant ladder, its rate function, and its tilt center - all of which the CONSTANT part of the coin leaves untouched (a pure detuning is invisible to every ratio $P(w)/P(-w)$ because ratios cannot see an overall shift of the exponent). The demon's fee therefore hides in the $\Delta\mu$ (which the variance/cumulant books of Ch.76/78 did see) and in this chapter's reshaping residual - the two ledgers Ch.75 billed, now separated into the constant and the conditional.

4. The 0/0 and the Constant

The 0/0 that closes this chapter: at $c = 0$ (reversible) and as the coin goes to pure detuning ($\delta \rightarrow \text{constant}$) the reshaping residual $\text{netR} \rightarrow 0$, so the ratio of reshaping-to-translation is 0/0. Its removable value is the State-Dependence Index of the bit: engaged ~6% detuned, harvest 19% - the fraction of the demon's work-curve action that cannot be absorbed by a shift and so must be understood as carefully-engineered conditioning, the part that a Maxwell-demon really spends.

Main Results

THEOREM

The Coin's Leading Effect Is a Constant: on the feedback round trip, the shifted mirror center satisfies $c = \Delta\mu$ for a pure detuning; measured engaged $c - \Delta\mu = -0.004$ (6%) - the coin is a translation to first order - while harvest deviates to $+0.025$ (19%) with an eightfold larger central reshaping residual (0.289 vs 0.037).

COROLLARY

The Split Is the State-Dependence of the Bit: the part of the demon's action beyond a constant detuning (netR) grows with the strength of the conditioning - engaged nearly pure constant, harvest strongly shaped - quantifying how far each coin is from a mere priced offset.

COROLLARY

The Mirror Is Blind to the Constant: a pure detuning leaves every ratio $P(w)/P(-w)$, every cumulant symmetry, and every tilt center unchanged; its fee shows only in $\Delta\mu$ (the Ch.76/78 ledgers) and in the reshaping residual of this chapter - the two accounts Ch.75 closed, now separated.

Conclusion

Six lenses on one equality ended in Ch.80 with the coin moving the mirror's center. This chapter measures HOW: the demon, to first order, is a constant - its whole action is a detuning that the mirror cannot see - and only its state-dependent tail reshapes the curve. The split (6% engaged, 19% harvest) is the information cost in its purest measured form: part priced, part engineered, and the 0/0 between them resolved at last.

References

- [1] Crooks, G.E. (1999). Phys. Rev. E 60, 2721 (detailed theorem).
- [2] Sagawa, T. & Ueda, M. (2010). Phys. Rev. Lett. 104, 090602 (information dependence).
- [3] Parrondo, J.M.R., Horowitz, J.M., Sagawa, T. (2015). Nat. Phys. 11, 131.
- [4] Touchette, H. (2009). Phys. Rep. 478, 1 (biased ensemble, tilt duality).
- [5] Book chain: Ch.74-75 (demon and coin), Ch.76 (TUR), Ch.77 (detailed mirror), Ch.78 (cumulant ladder), Ch.80 (tilt center).